

Anthony Mäkelä

Helsinki, FI & Gothenburg, SE | anthony.makela@gu.se

EDUCATION

University of Gothenburg

PhD in Mathematics

Gothenburg, SE

2025 - Present

University of Helsinki

Master's Degree in Mathematics & Statistics

Helsinki, FI

2023 - 2025

- Specialization: Geometry, Algebra and Topology

University of Helsinki

Bachelor's Degree in Mathematics & Statistics

Helsinki, FI

2020 - 2023

- Minor in Computer Science

RESEARCH & PUBLICATIONS

Negative Effective Divisors and Bridgeland Stability of Line Bundles on Surfaces. arXiv:2608.26080, 2026.

PROFESSIONAL EXPERIENCE

University of Gothenburg

Doctoral Researcher

Gothenburg, SE

2025 - Present

- Research in algebraic and complex geometry, with a focus on derived categories, Bridgeland stability conditions, and moduli spaces.

Ahlstrom

Senior Data Scientist

Helsinki, FI

2024 - 2025

- Developed neural networks, machine learning models and mathematical optimization methods for optimizing pulp-mill process recipes, using industrial data to support operational decision-making and improve production efficiency.
- Developed financial forecasting models using customized generalized additive models (GAMs) combined with autoregressive neural networks implemented in PyTorch.

UPM

Data Scientist

Helsinki, FI

2022 - 2024

- Developed hydrological forecasting models for Finland's inland water bodies and inflows to support hydropower production planning and spot-market energy trading decisions.
- Built PyTorch-based forecasting pipelines combining auto-regressive neural networks with traditional time-series methods to improve short- and medium-term inflow predictions.

University of Helsinki

Teaching Assistant

Helsinki, FI

2021 - 2023

- Teaching master's degree students in a course on Riemannian geometry, while also grading their weekly exercises.
- Teaching undergraduate students in a course on real analysis alongside vector analysis, while also grading their weekly exercises.
- Holding exercise sessions with the aim of supporting students in completing their weekly assignments.

Linear

AI Research Engineer

Helsinki, FI

2019 - 2021

- Developed deep-learning systems for extracting structured information from rasterized floor plans, including multi-task neural networks and segmentation and detection models.

- Developed predictive models for price fluctuations using gradient boosting and Bayesian methods.
- Developed NLP models for named-entity recognition and language models for automated description generation.

FinCap*Software Engineer***Helsinki, FI***2017 - 2019*

- Led the design and development of an application helping the internal sales team to publish, manage and sell newly developed real estate by the parent company. Reported directly to the CEO and the financial management.

PERSONAL PROJECTS

Tuple

- Single-handedly built Tuple, a Ruby on Rails web application for interactive learning aimed at Finnish high school students and applicants preparing for university entrance exams. Tuple offers structured courses, lessons, quizzes, practice exercises, progress tracking, payments, and AI-assisted learning features, combining modern web technologies with cloud-based integrations to provide an engaging study experience.

TECHNICAL SKILLS

Skills: Python, C++, Unix/Linux, Ruby, R, SQL, JavaScript, LaTeX**Tools/Frameworks:** PyTorch, TensorFlow, Django, Ruby on Rails, OpenCV, Open3D